Total No. of Questions : 10]

SEAT No. :

P3063

[Total No. of Pages : 3

[5354] - 550-B

B.E. Mechanical

DESIGN OF PUMPS, BLOWERS AND COMPRESSORS

(Elective - IV)

(2012 Pattern)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Use of calculator, steam table is allowed.***
5) ***Assume suitable data if necessary.***

Q1) a) Explain dimensionless parameters used in compressible flow machines. **[6]**

b) Explain performance characteristics of fan, blower and compressors. **[4]**

OR

Q2) a) Write a short note on stage velocity triangles. **[4]**

b) A double acting reciprocating pump running at 40 rpm. is discharging 1.0 m^3 of water per minute. The pump has a stroke of 400 mm. The diameter of the piston is 200 mm. The delivery and suction head are 20 m and 5 m respectively. Find the slip of the pump and power required to drive the pump **[6]**

Q3) a) Explain Indicator diagram in Reciprocating Pumps. **[4]**

b) A single acting reciprocating pump has a plunger diameter 250 mm and stroke of 450 mm and it is driven with S.H.M. at 60 r.p.m. The length and diameter of delivery pipe are 60 m and 100 mm respectively. Determine the power saved in overcoming friction in the delivery pipe by fitting an air vessel on the delivery side of the pump. Assume friction factor = 0.01 **[6]**

P.T.O.

OR

Q4) a) Explain various forms of corrosion occurring in hydraulic machines **[5]**

b) Explain general design consideration of Pump Casing **[5]**

Q5) a) What are main causes for noise generation? What are methods for reducing the fan noise? **[8]**

b) Explain the different losses in fans , blowers and compressors. **[8]**

OR

Q6) a) Explain the different fan stage parameters with usual notations. **[8]**

b) Explain in detail the design procedure, selection and optimization of blower. **[8]**

Q7) a) What is surging and stalling ? What are its effect ? How it is developed? **[8]**

b) A centrifugal blower with a radial impeller produces a pressure equivalent to 100 cm column of water. The pressure and temperature at its entry are 0.98 bar and 310 K. The electric motor driving the blower runs at 3000 rpm. The efficiencies of the fan and drive are 82 % and 88 % respectively. The radial velocity remain constant and has a value of 0.2 u_2. The velocity at the inlet eye as 0.4u_2 . If the blower handles 200 m^3/min of air at the entry condition determine

i) Power required by the electric motor

ii) impeller diameter

iii) Inner diameter of the blade ring

iv) air angle at entry

v) impeller widths at entry and exit

vi) number of impeller blades

vii) The Specific speed. **[10]**

OR

Q8) a) Explain function of an aerofoil and discuss characteristics curve of airfoil. **[8]**

b) An axial fan stage consisting of only a rotor has the following data :-

Rotor blade air angle at exit	10^0
Tip diameter	60 cm
Hub diameter	30 cm
Rotational speed	960 rpm
Power required	1 kW
Flow coefficient	0.245

(Inlet flow conditions P1 = 1.02 bar and T1 = 316 K). Determine the rotor blade angle at the entry, the flow rate, stage pressure rise, overall efficiency, degree of reaction, and specific speed. **[10]**

Q9) a) Explain Enthalpy-Entropy diagram of a compressor in detail. **[8]**

b) Explain the following terms

i) degree of reaction

ii) slip factor

iii) choking **[8]**

OR

Q10) a) Explain the performance characteristics curves of Centrifugal Compressor. **[8]**

b) What is work done factor for an axial flow compressor stage? How does it vary with no of stages? **[8]**